

Woolworths Online Order and Delivery System – Enterprise Systems Analysis, Design, and Modelling

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Table Of Content

Introduction	3
Organisational and System Analysis	3
Problem Statement	3
Proposed Solution	5
Organisation Overview	7
Departmental Structure and Responsibilities	7
Stakeholders and User Roles	9
Information from Stakeholders	10
System scope, boundaries, and assumptions	11
Requirements	11
Business Goals	12
User Requirements	12
Store Operation Staff should be able to:	13
Requirement UML Modelling	14
Initial Structural Design and Implementation	14
Conclusion	15
References	15
Appendices	15

Introduction

- ☐ Brief intro to the project
- ☐ Purpose of the report
- ☐ Overview of what each section covers

Organisational and System Analysis

- ☒ ~~2.1 Organisation Overview~~
- ☒ ~~2.2 Departmental Structure and Responsibilities~~
- ☒ ~~2.3 Problem Statement~~
- ☐ 2.4 Stakeholders and User Roles
- ☐ 2.5 System Scope, Boundaries, and Assumptions

Problem Statement

As online shopping continues to grow in popularity due to its convenience, Woolworths New Zealand has identified a need to implement an online system of its own (this is an assumption that Woolworths New Zealand does not have an online ordering and delivery system; the proposed idea is based on this assumption). Currently, Woolworths New Zealand does not have a dedicated online order and delivery system, which is a significant issue for multiple departments to perform effectively.

Customer service challenges

From the customer's perspective on the manual purchasing of groceries, there is no efficient way in which customers are able to compare and decide different product choices based on their nutritional value, price, and other product attributes. Customers would also not be able to access Woolworths

online, as there is no system that provides the ability for customers to purchase goods. The customer would not have the convenience or mobility issues to be able to travel to a supermarket.

From the standpoint of a customer service representative, the absence of an online platform implies that customer service would have to be conducted in person, in-store for refunds, and deal with customer complaints, as well as have no organised method to manage digital customer complaints. This makes it difficult for Woolworths New Zealand to quickly address, track, and solve issues. This would erode consumer loyalty and confidence.

Operational challenges

The challenges that the store operation employees faced were that they were not able to visualize the total number of products offered in the store, thus needing to manually record and manage product stocks, which could increase the chances of human error. Employees would not be able to have statistical data on which product has the most demand, which leads to poor positioning of where products should be stocked.

Shift managers have no digital tools to delegate tasks or monitor the progress of staff, relying entirely on in-person communication. Store managers have no central system or efficient way for the store manager to track the performance of the store, which would help determine potential operational bottlenecks. Currently, the store manager relies on manual reporting, which is time-consuming and a possible task that could involve human error.

Supply chain and logistics impact

This issue would also affect the supply chain and logistics department, as procurement specialists are unable to make the most effective decisions in purchasing and sourcing products due to a lack of live visibility on real-time demand data. This provides inaccurate data to the procurement specialist, which would also affect the inventory controller, as they monitor and maintain optimal stock levels across store locations based on the procurement specialist's data. This would risk overstocking or understocking across various locations.

This also impacts delivery drivers, as ineffective supply allocation across store locations makes it difficult to plan and fulfil deliveries efficiently, raising overall operating expenses.

Proposed Solution

Woolworths New Zealand needs to develop an Online Order and Delivery System that allows customers to place an online order, in which customers are allowed to browse and search for products online. The customer should also be able to add products to their shopping cart and be allowed to check out. Customers would also be able to save a shopping list to reuse previous lists of orders to make it faster to shop for weekly used products. Customers will be able to create an account so that their details and saved shopping list are stored on the account for ease of access to previous orders or frequently purchased products. Customer service representatives should have access to a digital complaint and inquiry management system that allows them to log, track, and resolve customer issues efficiently, improving response times and maintaining customer loyalty.

In the store operation department, the store operation employee would be able to view digital information on the online system about what orders to pick and pack to prepare for delivery. The Shift Managers would use the system to delegate task on the store operation employees and be able to monitor their progress. The store manager would be able to have a view of performance metrics (tracking orders received, processed, and fulfilled), which would help identify bottlenecks or delays. Other task that is needed to be able to be performed include inventory oversight, reviewing customer feedback, and managing store information (display of online promotions, announcements and opening hours).

For the supply chain and logistics department, the system should provide procurement specialists with access to real-time demand data, receiving messages about stock levels, especially if they are low, and track purchases made and the delivery of suppliers on the system. The procurement specialist should also be able to generate reports for planning future purchases. Inventory controllers should be able to monitor live stock levels across all store locations through the system, and monitor incoming stocks to

ensure inventory is correctly recorded. Just like the procurement specialist, the inventory controller would also be able to generate reports on inventory to see which products need restocking the most and the least.

Delivery drivers should be assigned schedules automatically based on their availability and location, and should be able to update delivery statuses in real time so that customers are notified of their order progress. Drivers should also be able to flag issues on the system that need urgency.

Organisation Overview

Woolworths New Zealand (rebranded from Countdown) is a subsidiary of Woolworths Group Limited, an Australian publicly listed company. Woolworths New Zealand is a major supermarket chain whose headquarters are located in Mangere, Manukau, Auckland. Woolworth Group was founded by Percy Christmas on September 22, 1924, in Sydney, Australia. It is one of the 3 major supermarkets (Pak'n Save, Woolworths, New World) in New Zealand. Currently, there are 185 supermarkets^[1] in New Zealand and 20,000 employees^[1] working in New Zealand.

Woolworth's core values are providing convenience, value, range, and quality, and a motto of "The Fresh Food People"^[2]. Thus, Woolworths needs dependable information systems to manage its operations across shops, distribution centers, and digital platforms because it depends greatly on customer satisfaction. One of these operations that would be focused on in this report is the Online and Delivery System, since it offers greater convenience and the ability to save time by ordering groceries online. These advantages in these systems align with the organisation's values, which are to provide the best experience for customers.

Departmental Structure and Responsibilities

In the ordering and delivery systems in Woolworths New Zealand, there are 4 critical departments in which forms the operation of the system. These systems are Store Operations, Supply Chain & Logistics, Customer Service, and IT.

- Store Operations

The store operations department for Woolworths New Zealand in 185 supermarkets^[1] focuses on restocking shelves and maintaining the food freshness of merchandise. In the context of an online ordering and delivery system, they are responsible for managing orders online by picking and packing them for delivery.

- Supply Chain & Logistics

The Supply Chain & Logistics department for Woolworths New Zealand focuses on managing stock movement from the supplier and distribution centres to the store. In the context of an online ordering and delivery system, they are responsible for assigning deliveries to complete orders and managing their delivery schedule, as 22%^[3] of orders online are fulfilled within 2 hours, and 57%^[3] of orders online are fulfilled within the same day of ordering, as well as ensuring there is enough stock for deliveries.

- Customer Service

The Customer Service department for Woolworths New Zealand focuses mainly on making sure that all customers are satisfied with the service and merchandise by handling customer inquiries, complaints, and issues. In the context of an online ordering and delivery system, this department is responsible for handling customer complaints and issues, assisting customers online, and processing refunds for incorrect or out-of-stock merchandise on the online ordering and delivery system.

- IT

The IT department for Woolworths New Zealand focuses on maintaining all digital platforms and technology systems, ensuring that user data is secure, developing improvements to the website, and resolving technical issues. In terms of the context of the online ordering and delivery system, the IT

department is responsible for maintaining the system, integrating inventory, supply chain, and payment systems.

- People and Culture (HR)

The People and Culture Department for Woolworths New Zealand focuses on making sure that all employees are well-trained, supported in their well-being, and manage payroll and recruitment. In the 2025 annual report, mental health programs such as 'Sonder' and leadership skills are completed by over 55,000^[3] employees. This department also manages wage agreements and ensures there is an inclusive environment and culture.

- Finance

The finance department for Woolworths New Zealand is involved in managing the budget, managing the cash flow, planning, and analysing financial decisions, analysing financial risk, as well as producing financial reports. These responsibilities are important in overseeing the financial report of Woolworths New Zealand. Food's total sales reached \$8,286 million^[3] with normalised sales.

- Marketing

The marketing department for Woolworths New Zealand's main goal is to build brand awareness, build brand loyalty through rewards, and research the trends of the market, creating promotions and advertisements. This includes the marketing department managing the grocery shopping website, which contributed to having 2.1 million^[3] active everyday reward members in Q4 of 2025.

Stakeholders and User Roles

Department	Stakeholder	Role
Store Operations	Store Operations Staff	Picks and prepares orders in the store
Store Operations	Store Manager	Oversees overall store operations and monitors system performance.
Store Operations	Shift Manager	Oversees store operations and staff delegation
Supply Chain and Logistics	Delivery Driver/Employee	Fulfils deliveries to customers
Supply Chain and Logistics	Procurement Specialist	Specialist Purchases stock for store locations
Supply Chain & Logistics	Inventory Controller	Manages stock distribution across store locations
Customer Service	Customer	Primary user of the online shopping platform
Customer Service	Customer Service Representative	Handles customer complaints and refunds

Information from Stakeholders

The methods used to gather information from stakeholders are mainly interviewing stakeholders with questions. As well as viewing similar systems in the same industry to find out what the existing common features are in the system.

Customer stakeholder questions (This response is from a friend acting as a customer stakeholder):

Question	Answer
What are some of the problems you have with the ordering process in existing systems from other organisations?	When ordering something, finding the product is hard. Sometimes the filter does not work. Adding items to the cart is slow. Item stock is not displayed properly (want real-time stock information).
How often do you currently shop for groceries, and what frustrates you most about the in-store experience?	The client said he went shopping once a week and that the thing that frustrates him the most is the checkout lines, which take too long.
What would make you choose online grocery shopping over visiting a store in person?	Prices should not be too expensive for online ordering, and people selecting items should be choosing quality products. The online experience should be nice and easy to use, such as not having too many unnecessary buttons (search bar, filter).
How would you prefer to be notified about your delivery status – SMS, email, or in-app notifications?	In app notifications, sms should be an option as another method of notifying.

Would access to exclusive online-only deals or Everyday Rewards points influence how often you shop online?	The stakeholder stated that if money is saved, filling out a form is not required. Then a deal or an everyday reward feature would be fine to have.
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Shift Manager stakeholder questions (This response is from me):

Question	Answer
How do you currently assign tasks to store operation staff, and what are the biggest challenges with this process?	Currently, the shift manager assigns tasks, having scheduled meetings daily on what each staff member's task is to complete, and the biggest challenge is having to organise a in person meet everytime since it is time-consuming.
How do you currently monitor the progress of staff during a shift, and how time-consuming is this?	To monitor the staff's progress, each staff will have a paper time sheet for each staff member, and when they are done with their shifts, they write down on that sheet. The sheets are collected at the end of every day to be evaluated by the shift manager.
What are the most common issues that arise during a shift that slow down operations?	Common issues are that it is hard to make any changes in plans, and you will need to meet in person to inform employees.

How are staff currently informed about which products need to be picked and packed for an order?	Staff are informed by getting a paper receipt on the board
How do you currently handle situations where a staff member falls behind on their tasks or is unavailable?	
What features would be most useful to you in a system that allows you to delegate and monitor tasks digitally?	
How would you prefer to view the progress of tasks – through a dashboard, notifications, or another method?	

System scope, boundaries, and assumptions

- Assumptions
- 1. Assuming Woolworths has not implemented an order and delivery system and still relies on in-person grocery shopping.

Requirements

- ☐ Clearly defined functional requirements
- ☐ Clearly defined non-functional requirements
- ☐ Traceability between stakeholders and requirements

Business Goals

- Customer Service
 - Improve customer convenience by providing an accessible online grocery shopping platform.
 - Enhance customer satisfaction through effective delivery communication.
 - Improve customer trust through an accessible online complaint and refund platform.
- Store Operations
 - Decrease the amount of in-person communication that shift managers and retail employees use.
 - Reduce human error in store operations through digital task and order management.
 - Increase stock management precision with real-time stock visibility.
- Supply Chain & Logistics
 - Use automated delivery scheduling to increase delivery efficiency.
 - Incorporate real-time delivery status updates on deliveries.
 - Improve forecasting accuracy by providing procurement and inventory staff with real-time data demands.

User Requirements

Customer should be able to:

- Browse and search for products online.
- Compare products based on nutritional value and price.
- Add products to a shopping cart and check out.
- Save a shopping list for reuse.
- Create an account to store their details and saved lists.
- View store promotions, opening/closing dates, and news.
- Track and receive notifications about their delivery status.

Store Operation Staff should be able to:

- View real-time stock levels and product inventory.
- Receive order information to pick and prepare orders for delivery.
- Access demand data to improve stock positioning.

Store Manager should be able to:

- Track overall performance.
- Track order fulfilment rates.
- Suggest operation bottlenecks from analyzing the overall performance.

A delivery driver should be able to:

- Get automatically assigned delivery orders.
- Update the delivery schedule when assigned to a delivery.

A procurement specialist should be able to:

- Access to real-time data on stock levels to make accurate purchasing decisions.
- Place orders for stock from suppliers and manage them.

An inventory controller should be able to:

- Monitor the stock level of one or more different locations, and contact the procurement specialist for

A customer service representative should be able to:

- Logging and managing customer complaints.
- Processing refunds for incorrect or out-of-stock items.
- Notifying customers of issues with their order.

Requirement UML Modelling

- ☐ Use Case Diagram covering full system scope
- ☐ Minimum of five fully dressed Use Case Descriptions
- ☐ Correct use of include, extend, and generalisation relationships

Initial Structural Design and Implementation

- ☐ Initial UML Class Diagram
- ☐ Mapping between UML classes and planned C# classes
- ☐ Identification of anticipated design weaknesses prior to OOP application
- ☐ OOP-based implementation in C# that strictly conforms to the UML models.
- ☐ In the Class diagram, provide reasons for each component by writing a short summary of each.

Conclusion

References

1. Woolworths. (2024). Online Supermarket: Online Grocery Shopping & Free Recipes at countdown.co.nz. Countdown NZ Ltd.
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2. O'Connell, J., & O'Connell, J. (2024, November 19). 1987 The Fresh Food People. Australian Food Timeline. <https://australianfoodtimeline.com.au/fresh-food-people/>
3. WOOLWORTHS GROUP LIMITED. (2025). Annual Report 2025.
<https://www.woolworthsgroup.com.au/content/dam/wwg/sustainability/reports/f25/Woolworths%20Group%20Annual%20Report%202025%20.pdf>

Appendices

- GitHub
(<https://github.com/ShawnLeeyz/Woolworths-Systems-Analysis-Design-and-Modelling-Project-I.git>)

